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FLOOR-MOUNTED CART SECUREMENT SYSTEM

Abstract

A floor-mounted securement system can be used to secure carts in a cargo compartment or container (e.g., in the front-to-back direction of a cargo container). The system can include various components, such as one or more logistics tracks positioned along the floor of the cargo compartment. In addition, the system can include a stopper that can releasably attach to the logistics track and that can be slid along the track to change positions of the stopper.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/559,559, filed Feb. 29, 2024 and entitled “FLOOR-MOUNTED CART-SECUREMENT SYSTEM.” The aforementioned application is hereby incorporated by reference in its entirety

BACKGROUND

[0002] Carts are often used to transport goods within cargo compartments, such as cargo containers that can be transferred from ships to trailer trucks. The use of such carts may reduce the amount of time needed to move good into the container. Some cargo compartments include wall-mounted securement systems to secure carts in position and reduce the likelihood of shifting during transport.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0003] Examples of a floor-mounted system for securing cars are described in detail below with reference to these figures.

[0004] FIG. 1 illustrates a perspective view of an example of a floor securement system in accordance with the present disclosure.

[0005] FIGS. 2A-B illustrate different views of a cargo compartment of a cargo container with carts and an example of a floor securement system in accordance with an aspect of the present disclosure.

[0006] FIG. 3 illustrates an example of a track in accordance with an aspect of the present disclosure.

[0007] FIG. 4 illustrates a side cross-sectional view from an end of the track of FIG. 3.

[0008] FIG. 5A illustrates a perspective view of a stopper assembly in accordance with an aspect of the present disclosure.

[0009] FIG. 5B is a partial cutaway view of the stopper assembly of FIG. 5A from a different perspective.

[0010] FIG. 6 is a perspective view of a stud of the stopper assembly of FIG. 5A.

[0011] FIG. 7 show a partially see-through view of the stopper assembly of FIG. 5A in engagement with the track of FIG. 3 in accordance with an aspect of the present disclosure.

[0012] FIGS. 8A-B show a cross-sectional view of the stopper assembly of FIG. 5A in different stages of engagement with the track of FIG. 3.

[0013] FIG. 9 illustrates another example of a stopper assembly in accordance with the present disclosure.

[0014] FIGS. 10A and 10B is are perspective views of an alternative example locking mechanism for a stopper assembly in accordance with the present disclosure.

[0015] FIGS. 11A and 11B are perspective views of an example stopper assembly with the alternative example locking mechanism of FIGS. 10A and 10B.

DETAILED DESCRIPTION

[0016] The present disclosure describes a securement system that includes one or more components that can be removably attached to the floor of a cargo compartment so that the carts are secured into position. In some examples, the carts can be secured in a longitudinal direction (e.g., in the front-to-back direction of a cargo container) to reduce shifting while the cargo compartment is in motion. The system is designed to be able to accommodate different sized carts that may be in different positions within the cargo compartment from trip to trip. Components of the system (e.g., logistics tracks) that are fixed to the floor are designed to minimize the risk of tripping.

[0017] Carts are often used to transport goods within cargo compartments, such as trailer truck

cargo compartments. The use of such carts may reduce the amount of time needed to bring goods into the cargo compartment. However, these carts need to be secured to prevent them from rolling, sliding, or tipping over when the cargo compartment is in motion (e.g., moved by ship, crane or truck). Securing the carts in place can itself be a time consuming exercise.

[0018] In conventional cargo compartment set-ups, straps (e.g., logistic straps) can to secure the carts. The straps are effective at securing the carts in place but can be quite inefficient in the loading and unloading process. For example, when not in used the straps may be stored in large bins, where they often become tangled and require time and effort to untangle them before use. Properly tightening and fastening the straps can also be time consuming.

[0019] Wall-mounted securement systems have been developed. However, they all require the use of some amount of space adjacent to the walls of the cargo compartment. This makes them incompatible with many carts, which, in order to maximize their cargo-carrying volume, are sized to extend from wall-to-wall within the cargo compartment.

[0020] While some conventional systems can include components that attach to the floor of the cargo compartment, they are associated with various drawbacks. For example, these conventional system are often not adjustable (e.g., fore-aft), and if they are adjustable, then they can require additional tools. This is problematic for cargo compartments where the positioning of one or more carts may vary from trip to trip, even if the carts have predetermined dimensions. For example, for trailers that transport food, the front section of the cargo compartment may be refrigerated, in which case an insulated divider may be used to isolate the refrigerated portion of the cargo compartment. The positioning of the insulated divider may vary depending on the size of the refrigerated section needed for a particular trip, which in turn affects the positioning of the carts that are lined up in the non-refrigerated section.

[0021] Thus there is a need for a cart securement system that secures carts in an adjustable fashion that is not mounted on the interior wall of the cargo compartment and is more efficient to use than conventional straps. The system must also be safe to use.

[0022] The floor securement system **100** is shown in FIG. **1**, which includes track **300** and a stopper assembly **500**, further described herein.

[0023] Turning now to FIG. **2A**, a top down cutaway view of the cargo compartment **10**, with a plurality of carts **50** shown fitting within the cargo compartment **10**. Cargo compartment **10** may be the inside of a shipping container, a refrigerated truck trailer, or other cargo space known in the art. Cargo compartment **10** has an internal length **12** and an internal width **14**. In examples, the left interior wall **16** and the right interior wall **18** are generally parallel with each other. Cargo compartment **10** also has a front end **19** and a back end **20**, with the back end **20** being configured to open and close, e.g., fitted with doors. FIG. **2B** illustrates a section of the cargo compartment **10** from the side, in which two carts **50** can be seen between the floor **24** and the ceiling **26**.

[0024] Tracks **300**, such as the one shown in FIG. **1** and FIGS. **3-4**, may be positioned along the floor **24**, running generally parallel to the left and right interior walls **16**, **18** of the cargo compartment **10**. In an example, each track **300** is a single piece that runs the length **12** of the cargo compartment **10**. In other examples, multiple segments of track **300** can be laid end-to-end serially to run the entire length **12** of the cargo compartment **10**, or only a part of the length of the cargo compartment **10**. In the example of FIG. **2A**, two tracks **300** are laid in parallel along a part of the length **12** of the cargo compartment **10**, but other configurations are contemplated to be within the scope of this disclosure. In some examples, the track or tracks **300** can be oriented in any direction relative to the cargo compartment, including g side-to-side or laterally, diagonally, or fore-aft/longitudinally. The tracks **300** can also run along the entire length **12** of the cargo compartment **10**.

[0025] As seen in FIGS. **2A-C**, one or more carts **50** can be positioned within the cargo compartment **10**. Each cart **50** depicted in FIGS. **2A-C** has a bottom **52**, which has a length **54** and a width **56**. Cart **50** may also have a shelving structure **58** attached to the top surface of the cart **50**,

or another structure for helping to organize or secure goods on the cart **50**. A plurality of wheels **60** are attached to the cart bottom **52**, which may be mounted on rotatable casters. The bottom **52** has a lower surface **62** (which faces the floor **24** of the cargo compartment **10**) and an upper surface **64**, on which goods can be placed. There is a clearance height **66** between the cargo compartment floor **24** and lower surface **62** of the cart bottom **52**, and a height **68** between the cargo compartment floor **24** and the upper surface **64**. Cart **50** has a width **56** that is slightly less than the internal width **14** of the cargo compartment **10**. In the example of FIGS. 2A-C, each cart **50** is approximately half the width **14** of the interior of cargo compartment, so that two carts **50** may be arranged in rows side-by-side within the cargo compartment **10**.

[0026] In between the carts **50**, stopper assemblies **500** are positioned on top of the track **300**. Each stopper assembly **500** is releasably secured to the track **300**, preventing the carts **50** from rolling in the front-to-back (e.g., forward or -backward) direction when they are locked into the track **300**. In other examples, one stopper assembly **500** may be used to prevent multiple carts **50** from rolling in the front-to-back direction. For instance, a first cart can be positioned so that it abuts the front wall (or other secure fixture) of the cargo compartment **10**, and a second cart can be positioned directly behind the first cart **50** so that they abut each other. A stopper assembly **500** is then secured to the track **300** so that it engages with the second cart on the opposite side of the first cart. The stopper assembly **500** thus prevents both the first cart and the second cart from moving in the backward direction, and the front wall of the cargo compartment prevents both carts from moving in the forward direction. It is readily appreciated that three or more carts can be similarly secured, or that a stopper assembly **500** in front of a first cart in a row of carts can serve the same stopping function in the forward direction as the front wall.

[0027] Turning back to FIGS. 3 and 4, each track **300** comprises a base **302** and an inverted T-shaped channel **320** that runs the length of the track **300**, with the bottom of the “T” forming a slot **322** running the length of the track defined by edges **324** of the base **302** on either side of the slot **322**. The base **302** has a bottom surface **304** that rests on the floor **24** of the cargo compartment **10**, and two side portions **306**, which each include a top surface **308**, a bottom surface **310**, and a connecting portion **312** that connects the side portions **306** to the base **302**. In the example shown in FIG. 4, the connecting portions **312** are angled relative to the side portions **306** so as to provide a smoother profile of the track **300** when it is attached to the cargo compartment floor **24**, minimizing their risk of becoming a tripping hazard. The track **300**, upon being secured to the floor **24**, has a height **314** above the floor **24**, which is less than the clearance height **66** of the cart bottom **52** with the floor **24**. The track **300** therefore does not impede the movement of the cart **50** in the front-back direction within the cargo compartment **10**.

[0028] As shown in FIG. 3, circular cut-outs **325**, each having a diameter **326** are spaced evenly along the slot **322**, with the centers of adjacent circular cut-outs **325** spaced a distance **328** apart from each other.

[0029] Turning now to FIGS. 5A-B, different views of the stopper assembly **500** are shown. Stopper assembly **500** comprises a stopper **510**, a housing **540**, and two studs **570**. The stopper **510** comprises a base **511** and a top portion **520** extending from the top surface **512** of the base **511**. In an example, the top portion **520** is a flange that projects from the top surface **512** of the base **511**, with a front surface **522** and a back surface **524**. Front projection **526** and back projection **528** extend from the top end **530** of top portion **520** in the front and back directions, respectively. As described in further detail below, the front projection **526** and back projection **528** can each engage with a cart to help prevent tipping of the cart during acceleration and deceleration of the cargo compartment **10**. In an example, the top portion **520** includes side flanges **532** connecting its edges **534** to the top surface **512** of the base **511**. Side flanges **532** (e.g., gussets) provide additional strength and stiffness to the top portion **520**. In examples, the top portion **520** is formed of a solid piece of material, but one of skill in the art can readily contemplate other configurations and constructions that may reduce the amount of material used for the top portion while providing

sufficient strength and stiffness.

[0030] The base **511** includes first and second legs **516**, **518**, which are configured to rest on top of either side portion **306** of track **300** (e.g., the terminal edges or “feet” of the legs **516**, **518** rest atop, and can slide relative to, the side portions **306** of the track **300**).

[0031] Near the center of the base **511** and adjacent the top portion **520** is a housing **540**. Plunger **550** is disposed within the housing **540** along with spring **560**. The plunger **550** has a plunger body **552** and a stem **553** with a free end **554** that extends out from opening **542** in the top of the housing **540**. In examples, a ring **556** is inserted within a hole adjacent the free end **554** of the plunger stem **553** to retain the plunger **550** within the housing **540** when the stopper assembly is not in use. The plunger body **552** is cylindrical with a diameter slightly less than the diameter of the circular cut-outs **325** of the track **300**. In examples, the plunger body **552** can include any shape that can mate into the openings in the track **300**.

[0032] The spring **560** biases the plunger **550** into an extended downward position. A force pushing up on the plunger body **552** or pulling upwards on the ring **556** sufficient to overcome the force of the spring can shift the plunger **550** into an elevated position within the housing **540**.

[0033] The stopper assembly further includes two studs **570** positioned on either side of the base **511**. As shown in FIG. 6, each stud **570** comprises a stud body **572** having a top end **574** and a bottom end **576**. A disc or head **578** extends out from the bottom end **576**. The disc **578** has a cross dimension (e.g., diameter) that is smaller than the cross dimension (e.g., diameter) of the circular cut-outs **325** of the track but greater than the width of the slot **322**. The stud body **572** is dimensioned to be less than the width of the slot **322**. The top ends **574** of the studs **570** may be threaded, allowing them to be fastened to the base **511** using conventional fasteners such as washers **588** and nuts **589**, or screwed directly into threaded holes in the base **511**. In other examples, other fasteners or fastening systems can be used, or the studs **570** may also be adhered, bonded or welded to the base **511**.

[0034] The stud axes **579** of the studs **570** are aligned with each other and the axis **559** of the plunger **550** to form a straight line when viewed from above. The two stud axes **579** are positioned so that they (and the centers of their respective discs **578**) are spaced apart at a multiple of the distance **D** between two adjacent circular cut-outs **325** of the track **300** (i.e., a multiple of the distance **328** shown in FIG. 3). In the example shown in FIG. 1 (and as seen in FIGS. 8A and 8B), they are spaced five (5) **D**s away from each other, with the plunger **550** positioned directly in between, so that the plunger's axis **559** is spaced 2.5 **D**s apart from each stud axis **579**.

[0035] Turning now to FIGS. 7A-B and 8A-8B, a stopper assembly **500** such as the one shown in FIG. 5A can be placed on top of a track **300** such as the one shown in FIG. 3, so that the studs **570** are each aligned with one of the track's circular cutouts. As shown in FIG. 8A, the plunger **550** is not aligned with one of the cutouts in this position. As shown in FIG. 8B, pushing down on the stopper assembly with thus cause the studs **570** to be inserted into the circular cut-outs **325** so that the disks enter the channel **320**, and the plunger, being pressed upwards by the track **300**, is moved into its elevated position. In FIG. 8C, the stopper assembly is shifted along the track until the plunger aligns with one of the circular cut-outs **325** and, biased by the spring **560**, is pushed into the cutout to lock the stopper assembly into place on the track **300** in the horizontal direction. The discs **578** of studs **570**, now being out of alignment with the circular cut-outs **325**, prevent the stopper assembly from disengaging with the track in the vertical direction. The legs **516**, **518** of the base **511** prevent the stopper assembly **500** from rocking laterally.

[0036] To move the stopper assembly **500** to a different position (or to remove the stopper assembly **500** from the track), a user can pull up on the ring **556** until the plunger **550** is disengaged from the circular cut-out **325**, and then the stopper assembly can be shifted along the length of the track to a desired position. If removal is desired, the stopper assembly **500** can be disengaged from the track **300**, once the studs **570** are aligned with the circular cut-outs **325** of the track, at which point the stopper assembly **500** can be lifted up off the track.

[0037] Turning back to FIGS. 1A-C, it will be apparent to one of skill in the art how a plurality of stopper assemblies **500** can be used with a track **300** attached to the floor **24** of a cargo compartment **10** to secure multiple carts **50** in the front-back direction. Because the top end **530** of the stopper assembly (when it is attached to the track **300**) projects over the top of the cart bottom **52**, the front projection **526** and back projection **528** can extend over the upper surfaces **64** of the cart bottoms **52**, which may provide additional protection against the carts **50** being rocked in the forward and backwards directions when the cargo compartment **10** is in motion. The size, configuration and orientation of the front projection **526** and the back projection **528** may be adapted to accommodate different sized carts. In an example, it is further contemplated that the front projection **526** and the back projection **528** may comprise adjustable parts that can be configured to engage with carts of different sizes, heights, etc. It can also be readily appreciated that multiple tracks **300** can be laid along and attached to the floor **24** so that multiple stopper assemblies **500** can be used to secure the same cart, which may be advantageous in situations where heavy loads are being transported.

[0038] The components of the present disclosure can be formed with known methodologies, such as die-casting or forming out of sheet metal. In an example, the stopper assembly **500** is manufactured by die-casting aluminum. In other examples, such as the stopper assembly shown in FIG. 9, components of the stopper assembly **900** can be formed out of bending, stamping, and/or machining sheet metal, and then affixed to each other in a separate step, e.g., by welding or bonding.

[0039] FIGS. 10A and 10B show an alternative to the plunger locking mechanism shown in the example of FIGS. 5A-B, with FIGS. 11A and 11B showing how a stopper assembly **500** with this alternative locking mechanism would engage with a track **300**. Instead of a plunger **550**, a rotating key **1000** may be rotatably housed within the housing **540** of the stopper assembly **500**. The housing **540** may be configured to allow the key to rotate about its axis **1030** in a limited arc. At one end of the key is a handle **1010** that allows a user to rotate the key **1000** around an axis **1030** from a first unlocked position **1101** to a second locked position **1102**. The opposite locking end **1020** of the key **1000** is configured so that in a first position it engages with a circular cut-out and in the second position is able to slide along the slot. In the example of FIGS. 10A and 10B, the locking end **1020** is in the shape of a cylinder having a diameter **1022** equal to or slightly less than the diameter of the circular cut-out **325**, with two opposing horizontal cylindrical segments removed so that two parallel flat surfaces **1024** are formed, with the width **1026** between the flat surfaces **1024** being slightly less than or equal to the width of the slot **322** of the track **300**. In an example, the handle **1010** is oriented so that its length is aligned with the flat surfaces **1024**, allowing the user to determine the orientation of the flat surfaces **1024** without having to see them. Alternatively a visual marking can be applied to the handle to provide a similar benefit.

[0040] As can be seen in FIG. 11A, in the first unlocked position **1101**, the flat surfaces **1024** of the locking end **1020** are aligned with the slot **322**, allowing the stopper assembly **500** to slide along the track **300** in either longitudinal direction. When the axis of the key **1000** is aligned with the center of a circular cut-out **325**, rotating the locking end **1020** into the second locked position **1102** (e.g., by applying a rotational force to the key via the handle **1010**) engages the circular edges **1028** of the locking end **1020** with circular cut out **325** and prevents the stopper assembly **500** from sliding along the track **300**.

[0041] As used herein, a recitation of “and/or” with respect to two or more elements should be interpreted to mean only one element, or a combination of elements. For example, “element A, element B, and/or element C” may include only element A, only element B, only element C, element A and element B, element A and element C, element B and element C, or elements A, B, and C. In addition, “at least one of element A or element B” may include at least one of element A, at least one of element B, or at least one of element A and at least one of element B. Further, “at least one of element A and element B” may include at least one of element A, at least one of

element B, or at least one of element A and at least one of element B.

[0042] This detailed description is provided in order to meet statutory requirements. However, this description is not intended to limit the scope of the invention described herein. Rather, the claimed subject matter may be embodied in different ways, to include different steps, different combinations of steps, different elements, and/or different combinations of elements, similar or equivalent to those described in this disclosure, and in conjunction with other present or future technologies. The examples herein are intended in all respects to be illustrative rather than restrictive. In this sense, alternative examples or implementations can become apparent to those of ordinary skill in the art to which the present subject matter pertains without departing from the scope hereof.

Claims

1. A system for securing a cart within a cargo compartment, comprising: a track, the track comprising: a track base with a bottom surface and a top surface; a slot in the top surface having a first length and a first width; and a channel in the track base having a second width that is greater than the first width; a stopper assembly configured to be removably engaged the track, comprising: a stopper base having a top and a bottom; a top portion extending from the top of the stopper base; a first stud extending from the bottom of the stopper base and configured to slidably engage the channel; and a locking mechanism disposed at least partially within the stopper base and movable from a first position to a second position.
2. The system of claim 1, wherein the locking mechanism in the second position prevents the stopper assembly from sliding along the track.
3. The system of claim 2, wherein the locking mechanism comprises a plunger.
4. The system of claim 3, wherein the plunger is slidable from the first position to the second position.
5. The system of claim 4, wherein the stopper assembly further comprises a biasing mechanism that biases the plunger into the second position.
6. The system of claim 5, wherein the biasing mechanism comprises a spring.
7. The system of claim 2, wherein the locking mechanism comprises a rotatable key.
8. The system of claim 1, wherein the first stud further comprises a channel-engaging portion that is greater than the first width and less than the second width.
9. The system of claim 8, wherein the track further comprises a plurality of cut-outs evenly spaced along the slot, each of the plurality of cut-outs being configured to allow the channel-engaging portion of the first stud to pass through one of the plurality of cut-outs and into the channel.
10. The system of claim 9, wherein each of the plurality of cut-outs are circular, with a center of each of the plurality of cut-outs being a first distance away from each adjacent cut-out.
11. The system of claim 10, wherein the channel-engaging portion comprises a disc-shaped end.
12. The system of claim 11, wherein the locking mechanism comprises a cylindrical end having a diameter that is greater than the first width and less than or equal to the second width.
13. The system of claim 12, wherein a center of the cylindrical end of the locking mechanism and a center of the disc-shaped end are spaced apart by a second distance that is not equal to a whole multiple of the first distance.
14. The system of claim 13, wherein the stopper assembly further comprises a second stud extending from the stopper base, the second stud configured to slidably engage the channel and comprising a second disc-shaped end.
15. The system of claim 14, wherein the center of the disc-shaped end and a center of the second disc-shaped end are spaced apart by a third distance that is a whole multiple greater than 1 of the first distance.
16. The system of claim 15, wherein the third distance is an odd whole multiple of the first

distance.

17. The system of claim 16, wherein the second distance is half of the third distance.

18. The system of claim 1, wherein the stopper base comprises at least one foot that is configured to rest upon the top surface of the track when the first stud is slidably engaged with the channel.

19. The system of claim 18, wherein the stopper base comprises a first foot and a second foot, the first foot resting upon the top surface of the track adjacent a first side of the slot and the second foot resting upon the top surface of the track adjacent a second side of the slot.

20. The system of claim 1, wherein the track is configured to be attachable to a floor of the cargo compartment.

21. The system of claim 3, wherein the top portion of the stopper assembly is configured to engage with the cart to prevent movement in at least one direction when the plunger is in the second position.

22. A stopper assembly configured to be removably engaged with a track comprising a track base with a bottom surface and a top surface, a slot in the top surface having a first length and a first width; and a channel in the track base having a second width that is greater than the first width; the stopper assembly comprising: a stopper base having a top and a bottom; a top portion extending from the top of the stopper base; a first stud extending from the bottom of the stopper base and configured to slidably engage the channel; and a locking mechanism disposed at least partially within the stopper base and movable from a first position to a second position.
